Table 7.9: Immunization for tetanus

Active immunization	Passive immunization	Combined prophylaxis
a. Tetanus toxoid (TT) b. DPT	Antitetanus serum (ATS) Human Antitetanus immunoglobulins (HIIG)	Tetanus toxoid in one arm and ATS or HTIG in another arm.

Tetanus develops by the contamination of wound with *Clostridum tetani* spores. The spores germinate in reduced oxygen tension (anaerobic environment).

↓

The vegetative cell grows and produces a potent neurotoxin called tetanospasmin. The toxin is absorbed from the site of its production and enters into the blood stream. It ascends to central nervous systems (CNS) through motor nerves.

↓

The toxin blocks synaptic inhibition in the spinal cord. It acts presynaptically.

↓

The toxin affects most of the voluntuary muscles in the body, causing muscle rigidity and spasma due to uncontrolled contractions.

↓

The first symptoms appears in head and neck because of shorter length of cranial nerves. Masseter muscles are first affected causing 'Lock Jaw'.

↓

In severe cases progressive spasm of the back or extensor muscles produces opisthotonos (extreme arching of the back) usually death occurs due to respiratory paralysis.

Flowchart 7.4: Pathogenesis of *Clostridium tetani*

(0.1–0.2 mg/kg) injection. Antibiotic therapy with penicillin or metronidazole should be done for a week or more.

7.7.8 Prophylaxis

It is done by the following methods, which are as follows.

a. Surgical prophylaxis: It aims at removal of foreign body, blood clots and damaged tissue in order to prevent anaerobic conditions favorable for the germination of spores.

b. Immunoprophylaxis: Tetanus is a preventable disease. Immune prophylaxis is of 3 types, which is given in the (Table 7.9).

Infobits

Clostridial Toxins As Therapeutic Agents

Botulinum toxin is the most poisonous substance known, is being used for the treatment of specific neuromuscular disorders characterized by involuntary muscle contraction. Since approval of Botulinum toxin (botox) by the FDA in 1989 for 3 disorders – Strabismus (crossing of the eyes), blepharospasm (spasmodic contraction of eye muscles) and hemifacial spasm (contraction of one side of the face).

In 2000, dermatologists and plastic surgeons began using Botox to eradicate wrinkles caused by repeated muscle contractions as we laugh, smile or frown.

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